

Question	Marking Guidance	Mark	Comments
04.1	1. Binding of first oxygen changes tertiary / quaternary (structure) of haemoglobin; 2. Creates / leads to / uncovers second / another binding site OR Uncovers another iron / Fe / haem group to bind to;	2	Ignore ref. to 'positive cooperativity' unqualified Ignore ref. to named bonds 1. Accept conformational shift caused 2. Reject ref. to active site
04.2	5.6 x 10 ⁶ (red blood cells per mm ³);; Award 1 max for one of 2.8 x 10 ⁴ (standard form but ignoring dilution) OR 5 600 000 (correct but not standard form) OR 5.6 x 10 ⁵ (failure to use depth of liquid on slide);	2	
04.3	1. To avoid dealing with parts of cells; 2. To avoid counting same cells twice / more than once; 3. To be consistent / get comparable results;	2 max	3. Accept more accuracy 3. Ignore reliability / repeatability / reproducibility / precision / validity
04.4	There are fewer white cells, so no need to dilute (further to see enough);	1	Accept converse of too few to see if greater dilution / at 200 times Do not accept ref. to numbers of red and white cells unqualified Ignore ref. to white cells larger
04.5	White cells have a nucleus (that stains but red cells do not);	1	Accept converse for red cells

